

SYNOPSIS

(All times UTC)

At about 0548 on 4 November 2005, the UK registered fishing vessel *Harvester* was engaged in pair trawling with the UK registered *Ocean Harvest* when she was in collision with the Norwegian registered supply and standby vessel *Strilmøy*. Before *Harvester* sank, all her crew were transferred to *Ocean Harvest*. *Strilmøy* was holed above the waterline in the forepeak tank, and suffered some denting to her starboard side. She was able to continue her voyage.

Harvester and *Ocean Harvest* were owned by their skippers, who were brothers. Each owned a half share of both vessels. They had been pair trawling together for the last 12 years, and regularly fished the area to the north-east of Peterhead. Both fishing vessels had two radars on board, and when fishing, one was used for lookout and the other for maintaining the correct distance from the other vessel of the pair.

Strilmøy was a new vessel, handed over to her owners on 26 August 2005. She was on a regular run, between Stavanger and Peterhead, calling at four oil rigs in the North Sea to deliver stores and pick up returns. She was fitted with a Voyage Data Recorder, and had two radars operating, as well as seven VHF radio sets available on the bridge.

At the time of the collision, the visibility had reduced to about 0.5 mile in fog. The pair trawlers had detected *Strilmøy* by radar at about 5 miles, and the radar plotting aid on *Ocean Harvest* showed that it would pass close ahead of the vessel but very close to *Harvester*. The radar plotting aid on *Harvester* showed that *Strilmøy* was on a collision course. The skipper of *Ocean Harvest* tried to call *Strilmøy* on the VHF radio, shortly before the collision, but received no response. The watchkeeper on *Harvester* started to alter course when *Strilmøy* was about 0.2 mile away, but could not take sufficient action in time to avoid the collision.

The impact woke *Harvester's* crew who, after checking the damage to the vessel, decided to abandon ship. Both of the liferafts were launched and the crew boarded the starboard liferaft, before being picked up by *Ocean Harvest*. On board *Strilmøy*, the engines were stopped and she remained in the vicinity, contacting the fishing vessels to exchange identities and offer assistance. The FRC from *Strilmøy* was launched, initially to check for damage to her own hull, and later, at the coastguard's request, to pick up any flotsam from the wreck, which included the EPIRB.

Harvester sank at 0720, and *Ocean Harvest* returned to Peterhead, where she berthed at about 2300 that evening. The master of *Strilmøy*, after assessing the damage to his vessel, decided that because the hole in the forepeak was above the waterline, and the ingress of water could be controlled with the vessel's pumps, he could safely continue the voyage to Stavanger.

A recommendation has been made to the International Chamber of Shipping and the International Support Vessel Operators' Association to encourage companies to reassess their written instructions with respect to lookout and radar use, and to ensure that the instructions are complied with at sea.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *STRILMØY*, HARVESTER AND ACCIDENT

Vessel details		<i>Strilmøy</i>
Registered owner	:	Simon Møkster Shipping AS
Manager(s)	:	Simon Møkster Shipping AS
Port of registry	:	Stavanger
Flag	:	Norway
IMO Number	:	9328546
Call sign	:	LMYV
Type	:	Supply and standby vessel
Built	:	2005
Classification society	:	Norske Veritas Classification A/S
Construction	:	Steel
Length overall	:	88.0m
Gross tonnage	:	3331
Engine power and/or type	:	4 x Caterpillar 3516B, 10197 Bhp,7604kW 4 x AVK Generators, 2281 kVA Diesel Electric propulsion
Service speed	:	15.2
Other relevant info	:	Forward Azimuth Thruster – 880kW 2 x Bow Thrusters - 880kW 2 x 2500kW 360° pods aft Fully enclosed bridge with sound reception equipment, located on the Monkey Island.

Vessel details

Harvester (PD 98)
(Ex *Margona* - Ex *Celestial Star*)

Registered owners	:	Mr J and Mr B Stephen
Manager(s)	:	Peterhead Fishermen Limited
Port of registry	:	Peterhead
Flag	:	UK
Type	:	Fishing Vessel
IMO Number	:	7328920
Callsign	:	GUBM
Built	:	1973 Aberdeen
Registered number	:	PD 98
Construction	:	Steel
Length overall	:	26.22m
Length registered	:	23.86m
Gross tonnage	:	154
Engine power and/or type	:	475.00kW

Accident details

Time and Date	:	0550 – 4 November 2005
Sunrise	:	0720
Location	:	59° 16'N 001° 52'E 130 miles north-east of Peterhead
Damage	:	<i>Harvester</i> was damaged and then sank; the bulbous bow of <i>Strilmøy</i> was holed.
Injuries	:	No injuries

1.2 BACKGROUND

1.2.1 *Strilmøy*

Strilmøy had been handed over to her owners on 26 August 2005, a little over 2 months before the collision. The master had been with the vessel during the final stages of the build and handover, and had assisted in designing the final layout of the bridge. She was on a charter to a major oil company to supply three oil rigs in the Norwegian sector of the North Sea, and two rigs in the UK sector. To serve this charter, the vessel would sail from Stavanger with stores for the Norwegian rigs. Having delivered the stores, she would then visit the rigs in the UK sector to pick up any return items. These would be taken to Peterhead, where the stores for the UK sector rigs would be loaded. On the return trip, the stores would be delivered to the UK sector rigs first, and then any returns picked up from the Norwegian sector rigs, before returning to Stavanger. The round trip was expected to take a week (see Figure 2).

The charter required there to be two navigating officers on the bridge when the vessel was operating close to the rigs. The vessel carried a master and three watchkeeping officers, as well as an AB/cadet, who had completed his nautical education, but was accumulating sea time to qualify for his first Certificate of Competency. The watchkeeping officers were two chief officers and a second officer, with the senior chief officer and second officer sharing the 12-6 watch, and the junior chief officer and the AB/cadet sharing the 6-12. The master would also be on the bridge when manoeuvring near the rigs. No additional lookouts were employed because it was expected there should be two navigating officers on the bridge all the time.

The main propulsion of *Strilmøy* consisted of two azimuth pods at the aft end of the hull. These pods could be rotated through 360° to provide thrust in any direction. Additionally, reverse thrust could be provided by changing the propeller pitch on each pod. A smaller 360° azimuth thruster was fitted forward, as well as two tunnel bow thrusters. Engine trials had concluded that the quickest method of stopping the vessel was to turn both main thrusters through 90°, such that they were both pushing “inwards”. Reversing the direction of the propeller pitch took much longer to stop the vessel.

1.2.2 Simon Møkster Shipping AS

Strilmøy was owned and operated by Simon Møkster Holdings. The company was established in 1968, initially running a fleet of coastal vessels, and later moving exclusively to services for the North Sea oil industry. The company is ISO 9001 compliant, and operates under an ISM document of compliance issued on behalf of the Norwegian Maritime Directorate; by Det Norske Veritas.

1.2.3 *Harvester* and *Ocean Harvest*

Harvester and *Ocean Harvest* (see Figure 3) were co-owned by two brothers, with each brother owning half the shares in both vessels. They employed a total of 9 crewmen: 4 crew (in addition to one brother acting as owner/skipper) worked each vessel, while the ninth man covered illnesses and holidays, and was the only man who worked on both vessels.

On the fishing grounds, the vessels would take alternate catches. The skipper of *Ocean Harvest* would be on watch while towing for *Harvester's* catch; at the same time his crew would be sorting their catch. The roles would then be reversed. This meant that one of the skippers was on watch at all times, while the towing watch was shared between the remaining four crew members on the other vessel. Since there was always one of the skippers on watch, he effectively took charge of both vessels, and made the decisions regarding course and speed.

Figure 3



Ocean Harvest

1.3 NARRATIVE

(All times UTC)

At about 2200 on Tuesday 1 November, *Harvester* and *Ocean Harvest* sailed from Peterhead. This was later than their planned Monday morning departure because *Ocean Harvest*'s fresh water tanks had been washed and treated. They headed for their usual fishing grounds about 130 miles north-east of Peterhead and were expecting to be out for about 10 days. Unusually, there were 6 men on *Harvester*. The vessel's engineer was on his first trip and the additional man was carried to help familiarise him with his duties.

Harvester and *Ocean Harvest* began pair trawling at about 1200 on Wednesday 2 November. The weather was good until the following night, when the wind dropped and it began to get hazy. By midnight, the visibility had reduced to less than a mile, the sea was calm with a low southerly swell.

Strilmøy sailed from Peterhead at 1910 on Thursday 3 November 2005, bound for the Beryl Field, where she was due at 0730 the following morning. On the bridge were the master, junior chief officer and the cadet. The master stayed on the bridge until about 2000, when he handed the con over to the junior chief officer. He went to bed shortly afterwards.

Harvester and *Ocean Harvest* were towing on north/south-bound legs, and at about 0200 they hauled in the gear having completed a north-bound leg. This was *Ocean Harvest*'s net, and when they shot away again at about 0230, the crew of *Harvester* went to bed, with the exception of the engineer, who took over for his first towing watch. In *Harvester*'s wheelhouse, one radar was switched between the 3 and 6 mile range scale and used for lookout, and the other radar was set on a 0.25 mile range scale to keep station with *Ocean Harvest*. *Ocean Harvest*'s skipper was also on watch, and the course for both vessels was set south at 2 to 2.5 knots. Visibility was reducing, and the skipper of *Ocean Harvest* and the engineer of *Harvester* kept contact with each other by radio. Once the catch from the previous haul was stowed, they were the only two crewmen awake on either vessel. Both vessels were displaying the lights for vessels of their size engaged in trawling. However, neither was using the optional searchlights directed at the other vessel of the pair, as allowed under Annex II of the COLREGS, for fishing vessels operating in close proximity. Neither were they sounding fog signals.

On *Strilmøy*, the two officers who kept the 12 - 6 morning watch, were relieved by the junior chief officer at 0445 (0545 ship's time).

Intermittent fog patches, which had been present at the beginning of the 12 - 6 watch had, by the time of the watch handover, developed into continuous fog. *Strilmøy* was equipped with a sound reception device, fitted to the vessel's monkey island, but this was not used as the crew considered that any exterior

sounds were masked by the noise made by the wind and funnel fans. The master was not informed that the visibility had reduced, and had left no standing instructions for the OOW to do so. No fog signal was being sounded. Shortly before the junior chief officer took over the watch, two small targets were noted on the radar, coming from the starboard side at a range of between 11 and 12 miles. These were not *Harvester* and *Ocean Harvest*, but were vessels of similar size, also pair trawling. *Strilmøy* was being steered by autopilot on a course of 045° (T) at a speed of 13.5 knots.

The main propulsion system was a diesel electric drive system which allowed for rapid manoeuvring under all operating conditions. Therefore, as required by Rule 19b of the COLREGS, *Strilmøy* had her engines ready for immediate manoeuvre. The two radars were operating, and the lights for a power-driven vessel of more than 50 metres long were displayed.

The chief officer was alone on watch. The AB/cadet who had been on watch the previous evening, acting as lookout, was unaware that he was required on the bridge for the morning watch, so had started work assisting in cleaning the accommodation. However, the master had told the AB/cadet that he would be required on watch when the vessel was at sea when she had left Stavanger for the current trip. The chief officer did not call him to the bridge for lookout duties, even though he spoke to him by telephone, asking him to bring the cargo manifest to the bridge.

Harvester and *Ocean Harvest* appeared on the radar at 0509. The VDR recording of the 3cm radar shows that the targets were not acquired using the ARPA. The true trails function of the ARPA was operating, and this showed the fishing vessels to be heading south at slow speed. There was no radar function in use to assess the risk of collision with these vessels.

The engineer on *Harvester*, and the skipper on *Ocean Harvest*, noted *Strilmøy* at about the same time, at about 5 miles. Both started tracking *Strilmøy* with their automatic plotting aids, and noted that she would pass ahead of *Ocean Harvest* but was on a collision course with *Harvester*. This was approximately 14 minutes before the collision. The two fishermen confirmed with each other that they had seen *Strilmøy*, and continued plotting her approach.

At 0547, the VDR on *Strilmøy* recorded a VHF call from *Ocean Harvest*, during which the skipper asked “supply vessel, can you see us?” Shortly afterwards, *Harvester*’s watchkeeper started to alter course to starboard to try and avoid the imminent collision. Thirty seconds after the VHF call, *Strilmøy* struck *Harvester* on the starboard side under her ‘A’ frame. The impact rolled *Harvester* heavily to port, waking the crew, and throwing those sleeping on the starboard side of the cabin out of bed. The crew checked the engine room and fish hold, and discovered water entering both spaces. The bilge pumps in the engine room and

fish hold were started, and these seemed to be coping initially with the flooding in the engine room. Further checks revealed a split in the hull in the fish hold below the waterline, and a second split in the bulkhead separating the fish hold and engine room. The impact had breached the starboard fuel tank, and fuel was mixing with the water in the bilges and entering the fish hold. The crew put on their immersion suits, and some also donned their lifejackets. Both liferafts were launched from the wheelhouse roof, and both inflated correctly.

Ocean Harvest's skipper saw the collision take place. He saw the lights of his brother's vessel heel to port, and then his view of *Harvester* was blocked by *Strilmøy*'s hull. A short while later, his brother called on a private VHF channel and confirmed that they were still afloat, that there were no injuries, but that the damage to the vessel was severe. Further conversation between the brothers included the decision to cut the fishing net adrift. Initially, *Harvester* was to cut her warps to save the net, but when the angle grinder failed to work, the gear was cut away from *Ocean Harvest*. Meanwhile, *Strilmøy* remained alongside *Harvester*. The low swell was causing *Harvester* to roll into *Strilmøy*'s hull, causing further damage. This also made it dangerous to enter the starboard liferaft, and *Harvester*'s skipper moved his vessel ahead to avoid further damage.

On *Strilmøy*, the OOW did not observe the target of *Harvester* on the radar, and first saw the vessel when she was estimated to be about 50 metres away from his vessel. This would have allowed the OOW only about 7 seconds to avoid the collision. His initial reaction was to attempt to alter course to starboard by adjusting the azimuth pods. However, because they were steering in autopilot, this action was blocked until an adjustment of greater than 60° was attempted. As the initial adjustment had not worked, the next option was to stop the vessel. This was most rapidly achieved by turning the azimuth thrusters through 90° so that they were opposing each other. This could stop the vessel in about 150m from full speed, so was insufficient to avoid the collision. The master arrived on the bridge 30 seconds after the collision, having been woken by the impact. Seeing *Harvester* alongside to starboard, and noting that her gear was still deployed and leading under his own vessel, he pressed the auto-stop for the azimuth pods to ensure that he caused no further problems by getting the wire wrapped around the thrusters. *Strilmøy* was temporarily unable to manoeuvre, and *Harvester* remained alongside. Because *Harvester* was no longer pulling the gear through the water, her fishing gear slowly cleared as the weight came off the nets. Once *Strilmøy*'s master could see that the gear was clear, he moved his vessel away by about 20 metres.

Two minutes after the collision, VHF radio contact was established between *Strilmøy* and *Ocean Harvest*. It was agreed that *Strilmøy* would remain nearby in case further assistance was required. *Strilmøy*'s FRC was made ready, but not launched, because the master was aware that *Ocean Harvest* was going alongside *Harvester* to rescue the crew.

The two fishing vessel skippers agreed that *Ocean Harvest* would transmit a “Mayday Relay” message. This was sent by MF and acknowledged by Shetland Coastguard at 0622. Contact was established between the coastguard and *Strilmøy* at 0642, and it was suggested that the vessel should proceed to Lerwick as the nearest port of refuge. At 0655, the ETV *Anglian Earl* was tasked to standby the casualty, and then to escort *Strilmøy* to Lerwick. However, *Strilmøy*’s master decided to continue to Stavanger, and because there was no pollution from his vessel, and the vessels were outside UK waters, there were no powers to insist that *Strilmøy* proceed to Lerwick.

By 0654, all the crew of *Harvester* had been evacuated to *Ocean Harvest*. At 0720, *Harvester* sank in position 59° 16.’2N 000°52.’2E. At 0746, *Strilmøy* was asked by Shetland Coastguard to use its FRC to pick up any debris, and to hand it to *Ocean Harvest*. This they did, and at 0833 all vessels left the area, *Strilmøy* to continue to Stavanger via the oil fields, and *Ocean Harvest* to Peterhead. At 0949, the ETV was released and returned to her station.

Ocean Harvest arrived in Peterhead at 2300 that evening.

1.4 ENVIRONMENTAL CONDITIONS

Rising and setting data for 4/11/05		Time (UTC)
Sunrise	:	0723
Sunset	:	1526
Moonrise	:	1135
Civil twilight	:	0638
Nautical twilight	:	0550

The weather at the time of the collision was as follows:

Wind	:	Calm
Sea	:	Calm
Swell	:	Less than 1m, southerly
Visibility	:	0.5 mile in fog.

The British Admiralty chart number 292 (North Sea Offshore Charts Sheet 1), which covers this area of the North Sea, has a tidal observation point (tidal diamond) situated in position 59°20’N 00° 49’E, which is within 5 miles of *Harvester*’s sinking position. The predicted tidal stream at this point at 0654 was 006°(T) at 0.5 knots.

It further states in Rule 19c:

Every vessel shall have due regard to the prevailing circumstances and conditions of restricted visibility when complying with the Rules of Section I of this Part.

As stated above, the Rules of Section I apply in any condition of visibility. These include the Rules concerning Lookout, Safe Speed, Risk of Collision and Action to Avoid Collision. The requirement of Rule 19c implies that, in restricted visibility, there should be additional attention paid to these aspects.

When the vessels were in sight of one another, then the Rules of Part B Section II applied. These eight rules include the detail of the action to be taken in crossing, overtaking, and head-on situations, as well as defining the action of the “stand-on” and “give-way” vessels. In the final moments before the collision, the rules of this section now applied. *Harvester* became a stand-on vessel, and *Strilmøy* was still a give-way vessel. For *Harvester*, Rule 17 (Action by the Stand-on vessel) now applied. This states in part:

“i) Where one of two vessels is to keep out of the way the other shall maintain her course and speed.

ii) The latter vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.”

To summarise, before the vessels could see each other, Rule 19 applied, and both of the vessels should have taken avoiding action. Once they could see each other, *Strilmøy* should still have taken avoiding action, and *Harvester* maintained course and speed. Once it became apparent to the watchkeeper on *Harvester* that *Strilmøy* was not taking any action, then he was required to take action to avoid collision by his manoeuvre alone.

Rule 35 of the COLREGS and STCW95 both require the sounding of fog signals when any vessel is operating in or near an area of restricted visibility. None of the vessels involved in this incident was sounding the required signals.

There was no requirement for the watchkeepers on the fishing vessels to have received any formal training in either watchkeeping or the COLREGS, provided that one properly certificated crew member was carried. This means that fishermen, with no formal qualifications or experience, may be required to stand watch on some fishing vessels. In order to improve the overall knowledge of basic watchkeeping issues and principles, Seafish is introducing a bridge watchkeeping training course for unqualified fishermen (at no charge) from March 2006.

1.10 LOOKOUT REQUIREMENTS

The regulations concerning lookout are given in the COLREGS Rule 5, and in the code for Standards Training and Certification of Watchkeepers 1995 (STCW95). The UK Maritime and Coastguard Agency gives further advice in the form of Merchant Shipping Notices (MGN 137 M+F, Lookout During Periods of Darkness and Restricted Visibility, MGN 202 M+F, Navigation in Fog). This additional advice was not applicable to *Strilmøy* since she was not a UK registered vessel. However, similar advice is given by the Norwegian Administration, and required by the Norwegian Maritime Act.

STCW 95 Section A VIII/2 part 3.1 describes the principles to be observed in keeping a navigational watch.

Section 12 states the responsibility of the officer of the watch for safe navigation of the vessel, and compliance with the COLREGS. Section 13 reinforces the requirements of rule 5 of the COLREGS, and further explains that lookout shall serve the purpose of:

Maintaining a continuous state of vigilance by sight and hearing as well as by all available means, with regard to any significant change in the operating environment;

Fully appraising the situation and the risk of collision, stranding and other dangers to navigation; and

Detecting ships or aircraft in distress, shipwrecked persons, wrecks, debris and other hazards to safe navigation.

Section 14 requires that:

The look-out must be able to give full attention to the keeping of a proper lookout and no other duties shall be undertaken or assigned which could interfere with that task.

Section 15 requires that the duty of helmsman and lookout are separate, and the helmsman cannot be considered to be the lookout except on small ships where there is an uninterrupted view all round. It also states that the officer in charge of the navigational watch may be the sole lookout in daylight with the proviso that on each occasion the situation has been carefully assessed and it is safe for this to happen.

STCW95 also states that the management company of a ship have a responsibility for ensuring that the obligations given in the code are given 'full and complete effect'. It also reinforces the requirement for lookout given in Rule 5 of the COLREGS.

MCA advice to fishing vessels on lookout responsibilities is explained in MGN 84(F) Keeping a Safe Navigational Watch on Fishing Vessels.

“4.5. Unfortunately it is not possible to rely on every give-way vessel to keep clear. It is therefore vital to monitor the movement of ALL traffic. Remember that a vessel engaged in fishing does not always have the right of way. In restricted visibility, even with gear extended, a fishing vessel has no special privileges.”

In February 2006, the MCA updated its advice on keeping a safe navigational watch, by issuing two Marine Guidance Notes. These were:

MGN 313 Keeping a Safe Navigational Watch on Fishing Vessels, and
MGN 315 Keeping a Safe Navigational Watch on Merchant Vessels.

1.11 INTERNATIONAL SAFETY MANAGEMENT CODE

The International Safety Management (ISM) Code requires that a company's safety management system should ensure compliance with mandatory rules and regulations. This is achieved by including in the safety management system, instructions and procedures to ensure the safe operation of ships, and protection of the environment in compliance with relevant international and Flag State legislation. In addition to these requirements, the Code requires that the company should establish procedures to identify, describe, and respond to potential emergency shipboard situations.

The owners of *Strilmøy* held an ISM Document of Compliance valid for the operation of offshore supply and standby vessels. Det Norske Veritas (DNV), on behalf of the NMD, had issued *Strilmøy* with an Interim Safety Management Certificate which was due to expire on 25 February 2006.

Short term Interim Safety management certification, usually valid for 6 months, is issued by, or on behalf of, the Flag State to allow new, or newly purchased vessels to commence trading and generate the evidence required during the subsequent full SMC audit to demonstrate that the ISM system is operating effectively.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

An analysis of the hours worked by *Strilmøy*'s OOW, and the rest he obtained leading up to the accident, indicated that fatigue was unlikely to have been a contributory factor. Similarly, although analysis of the hours of work and rest of the engineer lookout on *Harvester* determined there was a 'low risk' of fatigue, the lookout had detected the presence of *Strilmøy* at an early stage and had taken some action to try to avoid the collision. There is no indication that the actions or decisions taken by either of the crews were influenced by fatigue.

2.3 THE COLLISION

The collision was caused by the poor lookout being kept on *Strilmøy* and the late action taken by *Harvester*. *Harvester* was not noticed by the OOW until approximately 50 metres ahead of *Strilmøy*, by which time any action taken by *Strilmøy* was too late to avoid a collision. Action taken by *Harvester*, could have been taken earlier and been more substantial, however, incidents of close passing were a common occurrence to the crews of *Harvester* and *Ocean Harvest*, and an assumption was made that *Strilmøy* would eventually alter course. The action taken by *Harvester* resulted in a reduced angle at which the two vessels collided, and probably prevented the fishing vessel from being rolled over by the impact with *Strilmøy*.

2.4 LOOKOUT

Research by MAIB showed that over 58% of all the collisions and groundings investigated by the MAIB over a 5-year period to 2004, can be attributed to single-handed bridge watchkeeping.

2.4.1 *Strilmøy*

Strilmøy's OOW was alone on the bridge. It was dark and the visibility was poor. Both of these facts should have required an additional lookout on the bridge in accordance with STCW 95 Section 15. A lookout was available, and the master had expected the lookout to be on the bridge, although there were no written instructions to this effect.

With the OOW alone on the bridge, and with fog reducing the visibility to about 0.5 mile, one could expect a higher degree of alertness to the requirements of keeping a lookout as described in Rule 5 COLREGS. This would include sight, hearing, radar, etc. and would include the use of an additional lookout.

From the VDR recording of noise on the bridge it can be ascertained that there was activity on the bridge from when the watches changed to the time of the collision. This indicates that the OOW was awake and doing things. However, the recording shows that the activity was not just that which would be expected from someone keeping the watch. There are sounds recorded consistent with drawers being opened and closed, as well as with paper charts being unfolded and parallel rulers being opened and closed. Analysis indicates that the OOW's attention was not focussed on the task of lookout, but on some other task.

2.4.2 *Harvester* and *Ocean Harvest*

The towing watch on *Harvester* was being taken by the vessel's engineer for his first time. He had taken towing watches before, but not on this vessel. The routine the two vessels had adopted was that one of the skippers was on watch at all times when engaged in pair trawling. This meant that the on watch skipper took control of the fishing operation and co-ordinated the course and speed of both vessels. To assist the watchkeeper on *Harvester*, a plastic laminated sheet was attached to the bridge front. This gave details of when the skipper was to be called, such as when approaching land, or when in doubt. This would have been used when *Harvester* was on passage to and from the fishing grounds, when it was likely that neither of the skippers was awake.

Both the skipper of *Ocean Harvest* and the watchkeeper of *Harvester* had noted the presence of *Strilmøy* on the radar. They had confirmed by radio that each had seen the radar target, and were content to maintain course and speed. Although the presence of *Strilmøy* had been noted 15 minutes before the collision, no action was taken, even though the risk of collision was correctly assessed. It was not until about 20 seconds before the impact that *Harvester* went to starboard. This was in part due to the regular number of close quarters situations the crew of *Harvester* and *Ocean Harvest* had experienced while pair trawling in clear weather and restricted visibility. They had become accustomed to other vessels altering course when in very close proximity.

Restricted visibility meant that, under Rule 19, the pair trawlers, as well as the supply ship, should have altered course (and/or speed) to avoid collision. Neither did.

2.5 USE OF RADAR

The COLREGS describe in Rule 7(b) (Risk of Collision) that:

Proper use shall be made of radar equipment if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation of detected objects.

Radar alone will only tell the observer that there is a target on a certain bearing at a certain range. It is the monitoring of the target's movement that will indicate the possible risk of collision.

The radar in use on *Strilmøy* had a number of methods available with which the risk of collision could have been assessed. If an EBL had been lined up with the target, bearing change could have been monitored. Risk of collision is said to exist when the compass bearing of an approaching target does not change appreciably. Since the radar was gyro stabilised, the bearings from the EBL were compass bearings, and risk of collision could have been assessed. Changing the setting of the radar to display relative trails, while cluttering the screen with trails from all targets, would have indicated the approach of the target on a steady bearing.

By using ARPA, systematic observations are carried out automatically for the observer. The ARPA could be used either to manually acquire targets, or to set an auto-acquire zone to acquire any target appearing in the zone. Even if the radar was not frequently monitored, an alarm would have been set to sound when the target information showed that the TCPA and CPA had encroached within pre-set limits. In *Strilmøy*, the ARPA functions were not being used.

From the observation of VDR data from a number of recent incidents, MAIB has noted that many OOWs are only using the true trail function to display target information on radars. The true trail function will only indicate where a target is heading, and will not directly indicate any risk of collision. Assessing risk of collision from the true vector information requires the observer to estimate where his ship and the target will be at some future point, and decide whether or not this is likely to result in a close quarters situation. It is a very imprecise and inappropriate method of determining if risk of collision exists.

2.6 ACTION TO AVOID COLLISION

With the visibility at 0.5 mile, and the two vessels not in sight of one another, Rule 19 of the COLREGS applied (see Section 1.9). This details the actions to be taken by vessels not in sight of one another. At that stage, *Strilmøy* and the fishing vessels should have taken action to avoid collision. Neither side did.

Under Rule 35 of the COLREGS all three vessels should have been sounding fog signals, but were not.

In accordance with the COLREGS, once *Strilmøy* appeared out of the fog, *Harvester* became the stand-on vessel, since she was engaged in fishing and *Strilmøy* was a power-driven vessel, the Rules of Part B Section II now applying (Conduct of Vessels in Sight of One Another). Rule 17 (Action by Stand-on Vessel), meant that *Harvester* could have taken action once it became apparent that *Strilmøy* was not taking sufficient action. There was, however, little time to make this assessment, as the vessels were, by then, at very close range.

The COLREGS also give directions regarding Lookout (Rule 5), Safe Speed (Rule 6), Risk of Collision (Rule 7) and Action to Avoid Collision (Rule 8). These rules apply in any condition of visibility. They deal with the routines to be

adopted by the OOW: that he should be keeping a lookout by all means available; he should be proceeding at a safe speed adapted to the prevailing conditions; he should be assessing the risk of collision; and taking early and substantial action to keep well clear.

2.6.1 Actions by the pair trawlers

The method of fishing and the working routine adopted, had one skipper on watch, with the other vessel following his instructions. At the time of the collision, the actions of the pair were being co-ordinated by the skipper of *Ocean Harvest*, and the watchkeeper of *Harvester*, being on his first trip with the team, might have been expecting to be told exactly what to do.

The pair trawlers had four options: to increase speed, decrease speed, turn to port or turn to starboard. While any action taken sufficiently early would have avoided the collision, the four possible actions would have had varying effect:

- Increasing speed would not have been effective. The idea of increasing speed would have been to cross ahead of *Strilmøy*; this would have endangered the fishing gear. Since she was towing, and the increase in speed would have been slow, the effect would have been limited.
- Slowing or stopping the engines would have had a greater effect, because as soon as the power was reduced, the weight of the net would have pulled the vessel astern, effectively stopping the vessel, and increasing the distance ahead at which *Strilmøy* would pass. This would, however, increase the chance that the propeller would be fouled by the gear.
- Altering course at the time *Strilmøy* first appeared from the fog would have allowed insufficient time for the alteration to take effect, due to the slow speed and the fact that she was towing. An alteration of course to port, rather than starboard, would have been turning away from the net and trying to open the net further. This additional resistance would have further delayed and reduced the effectiveness of the action, as well as putting the vessel beam on to the approaching vessel. Any collision in this attitude could have rolled the vessel over, resulting in the loss of life. An alteration to starboard would have closed the net, and because of the reduced resistance from the net, the alteration of course would have been faster acting, and would more likely have resulted in a glancing blow.

The action decided on was to alter to starboard, but only when *Strilmøy* was very close. This, in part, was because they were fishing. They mistakenly (since it was restricted visibility) considered that they had the right of way. This is contrary to the advice given in MGN84(F) Keeping a Safe Navigational Watch on Fishing Vessels (see Section 1.10). Another possible reason for taking no action, was that in the past 2 to 3 years, they had become used to being passed at close range or for avoiding action, where taken, to be carried out at close range.

The autopilot was used, and *Harvester* turned through between 20° and 30° before the collision. If the steering had been in hand control, rather than in autopilot, hard-over helm could have been applied and a more significant alteration made before the collision. Any problem with the tow, caused by this action, could have been sorted out once the danger of collision had passed.

The watchkeeper on *Harvester* was following Rules 5, 6 and 7 of the COLREGS, but the action taken was not taken early enough to avoid collision.

2.6.2 Actions by *Strilmøy*

The VDR record of the collision shows that no action was taken on board *Strilmøy* until a few seconds before the collision. Although visibility was reduced by fog, the vessel was still maintaining a speed of 13.5 knots and a steady course of 045° (T). Had the two fishing vessels been noticed when they first appeared on the radar screen 38 minutes before the collision, an early alteration could have been made to result in the safe passing. They were not noticed, and no action was taken.

The VDR record does show that action was taken on board approximately 10 seconds before the collision. An initial attempt to alter course was not successful, and the OOW decided to stop the vessel. The most effective method of stopping was to turn both azimuth thrusters through 90°, such that they were operating athwartships in opposition to each other. This stopped the vessel in about 100m.

It is clear that the OOW on *Strilmøy* was not keeping a lookout, so risk of collision could not be assessed, and no action would therefore be taken to avoid collision. The action taken by *Strilmøy* was very late, and resulted only in a slightly reduced impact. Had an effective lookout been maintained, action could have been taken much earlier and collision with *Harvester* avoided.

2.7 MEDICAL

Strilmøy's OOW held a valid medical certificate, which he had obtained in May 2005. This was a requirement before starting his employment with Simon Møkster Shipping AS. The OOW wore glasses for reading, but had met the acuity standards without the need for wearing them. As a result of the collision, he had his eyes re-tested, because he felt that he could not see the radar screen clearly without his glasses. The doctor who re-tested his eyesight concluded that it was surprising that his eyesight problem had not previously been identified. His medical certificate was endorsed to require that he wears glasses when watchkeeping. The OOW is to undergo corrective surgery for his eyesight, at his own request. This surgery was suggested by the doctor, but was not essential.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following safety issues have been identified in the foregoing analysis. They are not listed in any order of priority.

1. There is no indication that the actions or decision-making of either of the crews was influenced by fatigue. [2.2]
2. The collision happened due to the poor lookout kept on *Strilmøy*, and the late action by *Harvester*. [2.3]
3. *Strilmøy* did not have a dedicated lookout on the bridge as required by STCW. [2.4.1]
4. The VDR record indicates that the OOW's attention on *Strilmøy* was not focussed on the task of lookout. [2.4.1]
5. Restricted visibility meant that, under Rule 19, the pair trawlers and the supply ship should have altered course (and/or speed) to avoid collision. Neither did. [2.4.2]
6. The OOW on *Strilmøy* was not using any radar plotting technique that could be considered a systematic observation of detected objects, as required by the COLREGS. He was not using ARPA. [2.5]
7. Neither vessel was using fog signals as required by Rule 35 of the COLREGS. [2.6]
8. *Harvester* could have taken earlier and more effective action. [2.6.1]
9. *Strilmøy* did not take any action to avoid collision, until it was too late. [2.6.2]
10. Detailed instructions concerning actions in fog, calling the master, and the requirements for bridge watchkeepers had not been produced and made available on board *Strilmøy* before the collision. [2.8]

SECTION 4 - ACTION TAKEN

4.1 SIMON MØKSTER SHIPPING AS

1. ISM procedure concerning bridge operations has been updated to ensure that:
 - the OOW informs the master of reduced visibility;
 - the requirements for duty operations are emphasised;
 - the routine for handing over the watch is formalised; and
 - the requirements with regard to the active use of radar are implemented.
2. Company circular letters have been sent to all vessels, announcing the implementation of the updated procedure, and stressing the routines for bridge watchkeeping operations. A management team is to present this campaign on board all vessels.
3. A personal letter has been sent to all masters in the fleet reminding them of their responsibilities to ensure the correct watch handover procedure is being carried out.
4. Has established a campaign to improve attitudes to rules and watchkeeping.
5. A dedicated book has been issued for documenting night standing orders.

4.2 PREVIOUS RECOMMENDATIONS BY MAIB

Thirteen safety recommendations have been made by the MAIB since January 2004 for accidents involving collisions between vessels. These are listed at **Annex 1**.